

Balancing Tourism, Environment, and Community Satisfaction in Juneau: A TRES Model Approach

Summary

Juneau, Alaska welcomed over 1.6 million cruise passengers in 2023, generating roughly \$375 million in revenue yet placing growing pressure on local resources and fragile glacial landscapes. In response, we develop our **TRES (Tourism–Revenue–Environment–Satisfaction)** model and integrate it with a **multi-objective** framework to balance tourism benefits against environmental and social costs. These combined models address multiple challenges, including detailed **expenditure planning**, robust **sensitivity analysis**, model **adaptation to other overtourism locations**, assessing **location-based measure importance**, and exploring a **Cold Spot Boost Factor** for promoting lesser-visited attractions.

In constructing the **TRES** model, we focus on three subsystems—**environment** (carbon emissions and glacier melting), **revenue** (tourist numbers and tax rates), and **resident satisfaction** (infrastructure, per capita income, and unemployment). By connecting these components through single-period equations, we capture how policy levers like visitor caps and investment allocations shape Juneau’s sustainability. Our core results show that limiting annual visitors to about 980,000–1,100,000, while modestly raising bed-tax rates to 7.5–8.1%, can reduce environmental impact by nearly 25–30% and preserve strong economic returns.

We then combine **goal programming (GP)** with a **genetic algorithm (GA)** to optimize these subsystems simultaneously. This multi-objective approach helps decision-makers prioritize different outcomes—environment-first, economy-first, or social-priority—revealing flexible policy mixes that still meet crucial targets. In each scenario, our **expenditure plan and feedback loop** ensure all tourism income is reinvested into environmental protection and infrastructure enhancements, thereby reinforcing both sustainability and community well-being. A subsequent **sensitivity analysis** highlights that tourist capacity and infrastructure spending exert the largest impact on overall results.

Finally, we extend our framework to other heavily visited destinations by recalibrating model parameters and constraints, demonstrating that **location-specific conditions** (e.g., urban congestion versus glacial ecosystems) can shift which policies matter most. In addition, we propose a **Cold Spot Boost Factor** to guide tourists to lesser-visited sites, reducing bottlenecks at high-traffic landmarks without sacrificing revenue. Through these adaptations, our methodology remains broadly applicable, ensuring that local authorities can tailor sustainable tourism strategies to their unique economic and environmental realities.

Keywords: TRES Model; Sustainable Tourism; Multi-Object Programming; Genetic Algorithm; Environmental Protection; Resident Satisfaction

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1 Introduction

1.1 Background Information

In 2023, Juneau, the capital city of Alaska, welcomed over 1.6 million cruise ship passengers to its spectacular glacial landscapes. While generating approximately \$375 million in revenue, tourism simultaneously places considerable strain on local resources and infrastructure.

Local residents are increasingly concerned about the constant crowds and worry that the growing carbon footprint from tourism is accelerating the melting of their glaciers, particularly the Mendenhall Glacier. So it is necessary to develop some sustainable tourism practices.

This paper examines the complex dynamics of sustainable tourism in Juneau, focusing on developing a comprehensive model that balances economic benefits with environmental preservation and community well-being.

1.2 Restatement of the Problems

To realize the balance between tourism income, local satisfaction and sustainable tourism, we can divide the problem into three parts as follows:

- **Task 1:** The primary task is to build a sustainable tourism model for Juneau, Alaska. The model will consider key variables like visitor numbers, overall revenue and stabilization measures, while identifying optimization factors and constraints. A comprehensive expenditure plan must be developed to show how additional revenue can be reinvested to promote sustainable tourism. Along with a sensitivity analysis to determine the most significant factors.
- **Task 2:** Examine how this model could be adapted to other overtourism affected destinations. Analyze how different locations affect which measures become most critical for that specific destination. Finally, strategies need to be explored for using the model to promote less-visited attractions and develop a better balance in tourist distribution.
- **Task 3:** Provide an actionable memo to Juneau's tourist council by presenting evidence-based predictions, assessing the impact of various measures and offering strategic optimization guidelines.

1.3 Model Overview

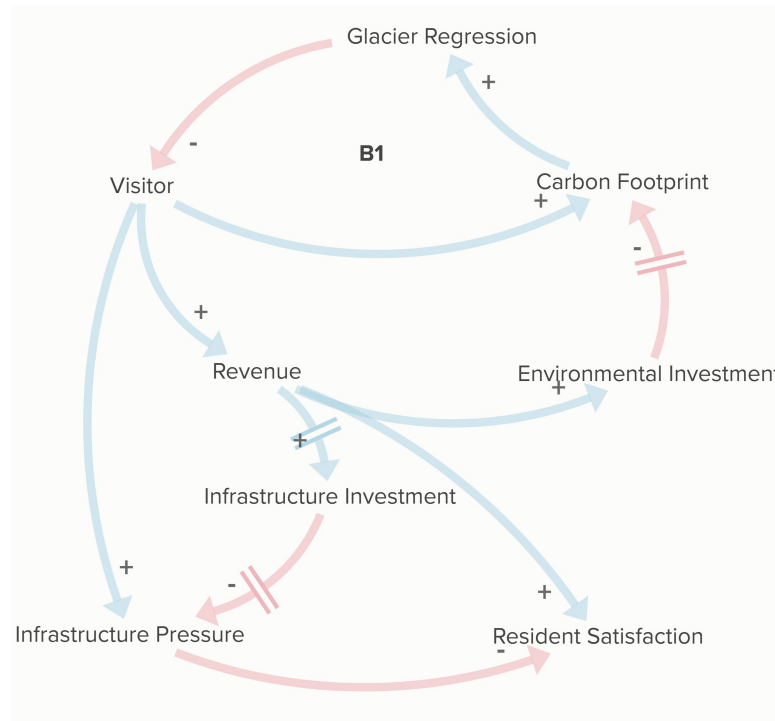


Figure 1: Dependencies of Our Model[1]

2 Assumptions

1. **Revenue Equivalence:** The total revenue (R) equals the tax income collected from tourism activities:

$$R = \text{TaxRate} \times \text{TourNum} \times \text{AverageSpending}$$

where **TaxRate** is the bed-tax rate, **TourNum** represents the number of tourists, and **AverageSpending** denotes the typical expenditure per tourist.

2. **Complete Revenue Investment:** All revenue (R) is allocated entirely to investments, distributed between Environmental Investment (I_{env}) and Infrastructure Investment (I_{infr}):

$$R = I_{\text{env}} + I_{\text{infr}}$$

where the allocation ratio is determined by the optimization model to achieve the best balance among environmental protection, economic benefits, and resident satisfaction.

3 Data Source

Table 1: Data Source Websites

Website	Usage
https://www.ncei.noaa.gov/access/monitoring/ao/	Acquire ClimateIndex
https://www.census.gov/programs-surveys/acs/data/data-via-ftp.html	Acquire Poverty Rate
https://www.bls.gov/lau/	Acquire Unemployment

4 Notifications

Table 2: Key Symbols and Their Descriptions in this Paper

Section	Symbol	Description
5.1 Environment	E	Carbon emission.
	G	Glacier melting rate (index of ice mass loss).
	TourNum	Tourist number.
	I_{env}	Environmental investment.
	ClimateIndex	Arctic Oscillation index (external climate driver).
5.2 Revenue	R	Total tourism revenue in a single period.
	TourNum	Number of tourists visiting the region.
	TaxRate	Bed-tax rate applied to visitor spending.
5.3 Satisfaction	S	Resident satisfaction.
	I	Per capita income.
	I_{infr}	Infrastructure investment.
	$I_{\text{infr,base}}$	Baseline level of infrastructure provision.
	TourNum _{max}	Reference capacity for tourist numbers.
	U	Unemployment level (or rate).

5 TRES (Tourism–Revenue–Environment–Satisfaction) Model

To address Juneau’s tourism challenges, we developed the **TRES** (Tourism–Revenue–Environment–Satisfaction) model - a comprehensive framework that examines four key aspects of sustainable tourism. The model helps decision-makers balance visitor numbers and tax policies while considering their impacts on the environment, local economy, and community well-being.

At its core, TRES uses **goal programming** to optimize four primary objectives(**factors we are optimizing**): minimizing environmental impacts (carbon emissions E and glacier melting rate G), maximizing tourism revenue (R), and maintaining resident satisfaction (S).

These factors are subject to several practical constraints(**factors serve as constraints**): tourist numbers (**TourNum**) are limited between 800,000 and 1,300,000 visitors annually, tax rates (**TaxRate**) range from 5% to 10%, and the allocation of investment between infrastructure and environment ($I_{\text{infr}}/I_{\text{infr,base}}$ and $I_{\text{env}}/I_{\text{infr,base}}$) must sum to 100% while individually staying within 20%-80%. By treating these factors as interconnected rather than isolated concerns, TRES offers a practical tool for developing **sustainable tourism strategies**.

The model works by analyzing a **single time period**, making it easier to understand immediate cause-and-effect relationships between different policy choices and their outcomes. Before diving into the specific equations, it’s worth noting that our approach relies on both **historical data** and **expert knowledge** to estimate key relationships, ensuring the model reflects real-world conditions as accurately as possible.

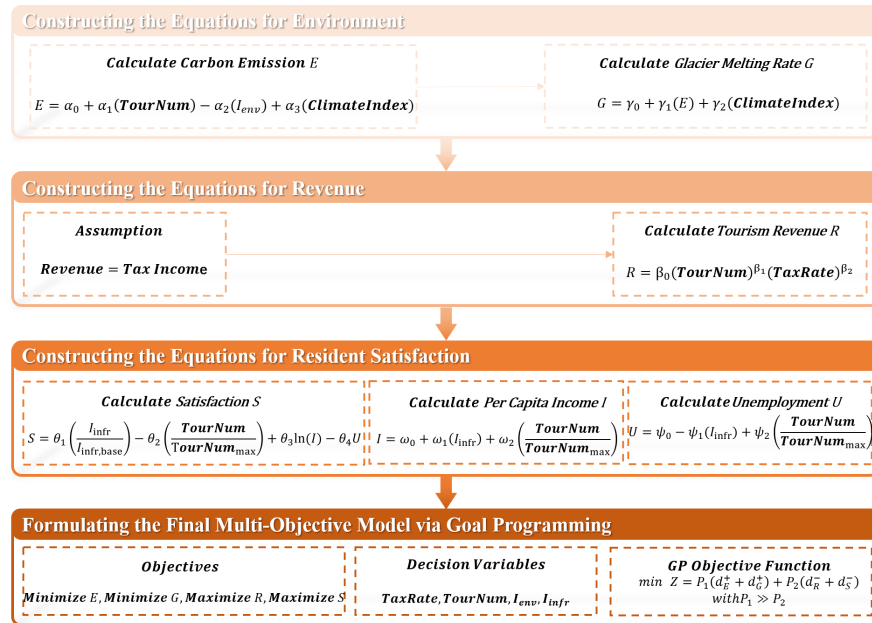


Figure 2: Model Setup

5.1 Constructing the Equations for Environment

In this section, we will take the Arctic Oscillation index as our **Climate Index** and introduce two key variables to represent the **environmental** state within a **single-period** framework [2]. Specifically, we focus on **carbon emission** E and **glacier melting rate** G , each formulated as a discrete (single-stage) function of its influencing factors. We write:

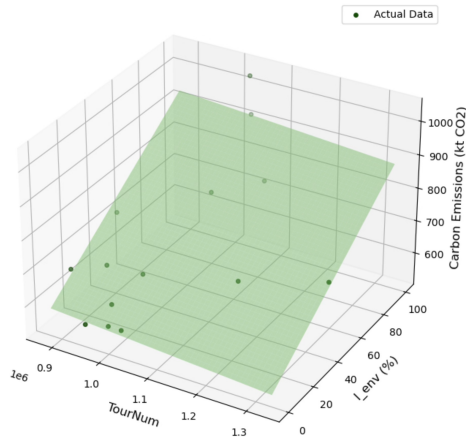
$$E = \alpha_0 + \alpha_1 (\text{TourNum}) - \alpha_2 (\text{I}_{\text{env}}) + \alpha_3 (\text{ClimateIndex}), \quad (1)$$

$$G = \gamma_0 + \gamma_1 (E) + \gamma_2 (\text{ClimateIndex}). \quad (2)$$

Here, $\alpha_1 > 0$ reflects how **TourNum** (*tourist number*) positively contributes to total carbon emission, while $\alpha_2 > 0$ captures the negative feedback of **environmental investment** I_{env} on emissions (i.e., the more we invest, the more pollutants can be offset). We also incorporate a **ClimateIndex** term via α_3 and γ_2 to represent external warming factors arising from the Arctic Oscillation. In the second equation, $\gamma_1 > 0$ indicates that as E (carbon emission) grows, the **glacier melting rate** G will likewise increase, linking large-scale warming trends to local glacier responses.

To implement these relationships, we rely on a combination of **historical data** (e.g., 2005–2019 records of carbon emission and glacier melting rates) and **expert judgment**. Concretely, one can perform a regression of (E, G) on $(\text{TourNum}, \text{I}_{\text{env}}, \text{ClimateIndex})$ to estimate the coefficients $\{\alpha_i, \gamma_i\}$, ensuring they respect our sign assumptions (such as $\alpha_1 > 0$, $\alpha_2 > 0$, $\gamma_1 > 0$). By embedding these discrete equations in a **single-period** model, we capture how tourism and environmental investment jointly affect both emissions and glacier melting without introducing continuous time variables, yet still reflecting real-world observations on how climate factors and human interventions shape environmental outcomes.

3D Fit: Carbon Emissions = f(TourNum, Environment Investment)



Fit: Glacier Changes = f(Carbon Emissions)

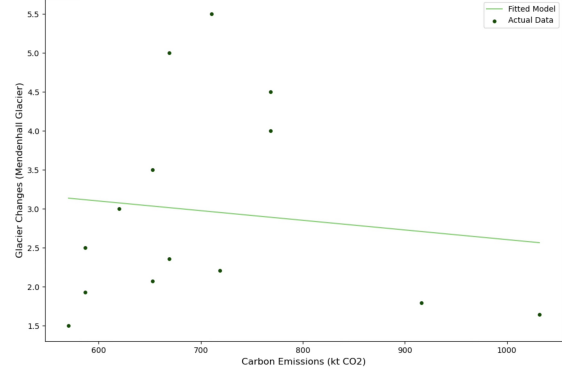


Figure 3: Environmental Impact Analysis

In Figure 3, there are (a) 3D visualization showing the relationship between tourist numbers (**TourNum**), environmental investment (I_{env}), and carbon emissions (E); (b) The relationship between carbon emissions (E) and glacier changes (G), with both fitted model and historical data points indicating the environmental impact.

5.2 Constructing the Equations for Revenue

In this section, we adopt a **single-period** perspective to capture how tourism revenue (R) arises from two key **decision variables**: the number of visitors (**TourNum**) and the bed-tax rate (**TaxRate**). Rather than modeling revenue over continuous time, we focus on an *immediate* or *static* assessment of how these factors jointly determine income within one period.

A first step is to recognize that **TourNum** is expected to have a *positive* influence on total revenue, reflecting standard economic logic that more tourists contribute more spending. Meanwhile, the bed-tax rate directly influences revenue, as a higher tax rate increases the amount of revenue generated per visitor.

To represent these combined effects, we introduce the following *algebraic* relationship:

$$R = \beta_0 (\text{TourNum})^{\beta_1} (\text{TaxRate})^{\beta_2} \quad (3)$$

where $\beta_0 > 0$ is a baseline scaling parameter for revenue, $\beta_1 > 0$ captures the degree to which additional tourists amplify earnings, and $\beta_2 > 0$ reflects how strongly the tax rate enhances revenue.

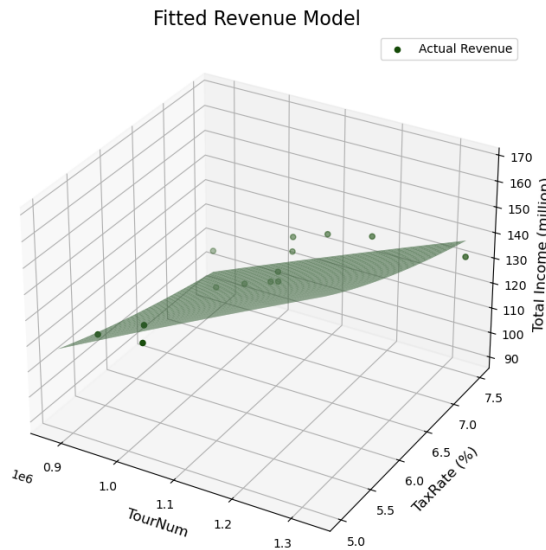


Figure 4: Visualization of Revenue Model

The surface in Figure 4 shows how tourism revenue (R) varies with tourist numbers (**TourNum**) and tax rate (**TaxRate**), while the scattered points represent actual historical data.

In practical applications, β_0 , β_1 , and β_2 can be estimated by fitting historical data on **TourNum**, **TaxRate**, and observed tourism revenue. If data are sparse or inconsistent, practitioners may rely on *expert assessments* (e.g., from local hotel associations or regional tourism boards) to validate or approximate these parameters.

By retaining this *single-period* format and avoiding any time-dependent terms, we ensure that the revenue equation can be integrated seamlessly with the **environmental subsystem** (Section 5.1) and other modules under a unified static framework, facilitating multi-objective analyses (e.g., balancing economic gains with environmental constraints) without resorting to differential equations or other dynamic modeling approaches.

5.3 Constructing the Equations for Resident Satisfaction

In this subsection, we adopt a **discrete**, single-period perspective to represent how local factors affect *Resident Satisfaction* (S)[3]. We write:

$$S = \theta_1 \left(\frac{I_{\text{infr}}}{I_{\text{infr,base}}} \right) - \theta_2 \left(\frac{\text{TourNum}}{\text{TourNum}_{\text{max}}} \right) + \theta_3 \ln(I) - \theta_4 U \quad (4)$$

where $I_{\text{infr}}/I_{\text{infr,base}}$ captures the relative level of infrastructure (e.g., public facilities or services) compared to a *baseline* standard, and $\text{TourNum}/\text{TourNum}_{\text{max}}$ measures how much crowding or congestion occurs (with $\text{TourNum}_{\text{max}}$ being a reference capacity). Here, θ_1 indicates how higher infrastructure provision *raises* satisfaction, whereas θ_2 reflects how crowding *depresses* it. Next, I denotes *per capita income*, entering in a logarithmic form to capture diminishing returns (i.e., boosting S , but at a decreasing rate); θ_3 governs that marginal effect. Finally, U is the unemployment level (or rate), and θ_4 reflects its negative impact on resident well-being.

Per Capita Income (I)

To facilitate this model, we express *per capita income* as a simple **discrete** equation influenced by local infrastructure investments and tourism activity. Specifically, we define:

$$I = \omega_0 + \omega_1 I_{\text{infr}} + \omega_2 \left(\frac{\text{TourNum}}{\text{TourNum}_{\text{max}}} \right) \quad (5)$$

where ω_0 is a bias or intercept term, ω_1 captures the effect of infrastructure investment (I_{infr}) on per capita income (e.g., through tourism-related employment), and ω_2 represents the amplification effect of tourist numbers on the economic scale (i.e., how $\text{TourNum}/\text{TourNum}_{\text{max}}$ enhances income levels).

Unemployment (U)

Similarly, we adopt a **single-period** form for unemployment, reflecting the idea that infrastructure investments and tourist activity influence joblessness. The revised discrete specification is:

$$U = \psi_0 - \psi_1 I_{\text{infr}} + \psi_2 \left(\frac{\text{TourNum}}{\text{TourNum}_{\text{max}}} \right) \quad (6)$$

where ψ_0 is an intercept capturing baseline unemployment, ψ_1 indicates how increased infrastructure investment (I_{infr}) reduces unemployment by creating jobs, and ψ_2 captures the potential temporary increase in unemployment due to seasonal fluctuations in tourist numbers (e.g., layoffs during off-peak seasons).

By combining Equations (4)–(6), we obtain a succinct *algebraic* subsystem for Resident Satisfaction in a **static** setting. In practice, one may *estimate* the parameters $\{\theta_j, \omega_k, \psi_\ell\}$ (for $j = 1, \dots, 4$, $k = 0, 1, 2$, and $\ell = 0, 1, 2$) by analyzing local data, such as surveys on residents' quality-of-life perceptions, government budget records on I_{infr} , **TourNum** data, and official unemployment reports. If direct historical data are lacking, *expert opinion* (e.g., from economic development committees or academic studies) can be used to approximate these coefficients. In this way, we preserve the *single-period* simplicity needed to integrate seamlessly with other modules (e.g., Environment or Revenue) while still capturing how infrastructure investment, tourism activity, income, and unemployment jointly shape overall resident satisfaction.

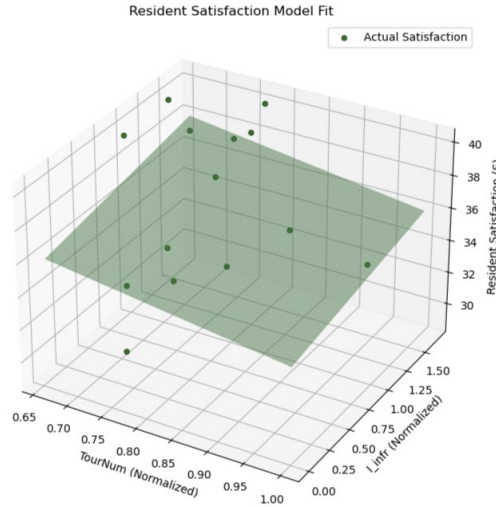


Figure 5: Three-dimensional visualization of the resident satisfaction model

The surface in Figure 5 shows how resident satisfaction (S) varies with normalized tourist numbers ($\text{TourNum}/\text{TourNum}_{\text{max}}$) and infrastructure investment ($I_{\text{infr}}/I_{\text{infr,base}}$), while the scattered points represent actual survey data. The model demonstrates that while both factors influence satisfaction, there exists an optimal balance between tourism intensity and infrastructure development.

5.4 Formulating the Final Multi-Objective Model via GP and GA

In this section, we *implement* the discrete Environment, Revenue, and Resident Satisfaction equations into a unified **goal programming** (GP) framework[4]. Then employ a **genetic algorithm** (GA)[5] to solve it. The ultimate goal is to minimize carbon emission (E) and glacier melting rate (G), while maximizing tourism revenue (R) and resident satisfaction (S). We use four decision variables:

$$\text{TaxRate}, \quad \text{TourNum}, \quad \mathbf{I}_{\text{env}}, \quad \mathbf{I}_{\text{infr}},$$

with all other variables and parameters defined in (7)–(12).

Step 1: Subsystem Equations (Discrete Form)

Recall the single-period relationships that produce E , G , R , and S :

$$E = \alpha_0 + \alpha_1 \text{TourNum} - \alpha_2 \mathbf{I}_{\text{env}} + \alpha_3 \text{ClimateIndex}, \quad (7)$$

$$G = \gamma_0 + \gamma_1 E + \gamma_2 \text{ClimateIndex}, \quad (8)$$

$$R = \beta_0 (\text{TourNum})^{\beta_1} (\text{TaxRate})^{\beta_2}, \quad (9)$$

$$S = \theta_1 \left(\frac{\mathbf{I}_{\text{infr}}}{\mathbf{I}_{\text{infr,base}}} \right) - \theta_2 \left(\frac{\text{TourNum}}{\text{TourNum}_{\text{max}}} \right) + \theta_3 \ln(I) - \theta_4 U, \quad (10)$$

where I (per capita income) and U (unemployment) are given by:

$$I = \omega_0 + \omega_1 \mathbf{I}_{\text{infr}} + \omega_2 \left(\frac{\text{TourNum}}{\text{TourNum}_{\text{max}}} \right), \quad (11)$$

$$U = \psi_0 - \psi_1 \mathbf{I}_{\text{infr}} + \psi_2 \left(\frac{\text{TourNum}}{\text{TourNum}_{\text{max}}} \right). \quad (12)$$

These six equations serve as constraints (or definitions) within the final optimization model, linking the decision variables $\{\text{TaxRate}, \text{TourNum}, \mathbf{I}_{\text{env}}, \mathbf{I}_{\text{infr}}\}$ to the key performance outputs E, G, R, S .

Step 2: Goal Targets and Deviation Variables

To fit the standard goal programming approach, we specify target values (limits or minimum thresholds) for each objective[4]:

- $E \leq E_{\text{limit}}$ (upper bound on carbon emissions),
- $G \leq G_{\text{limit}}$ (upper bound on glacier melting rate),
- $R \geq R_{\text{target}}$ (lower bound on desired tourism revenue),

- $S \geq S_{\text{target}}$ (lower bound on desired resident satisfaction).

We introduce ****deviation variables**** $d_E^+, d_E^-, d_G^+, d_G^-, d_R^+, d_R^-, d_S^+, d_S^-$, which measure any overshoot or shortfall relative to these targets. Hence, each goal is written in an equality form:

$$E(\mathbf{x}) + d_E^- - d_E^+ = E_{\text{limit}}, \quad (13)$$

$$G(\mathbf{x}) + d_G^- - d_G^+ = G_{\text{limit}}, \quad (14)$$

$$R(\mathbf{x}) + d_R^- - d_R^+ = R_{\text{target}}, \quad (15)$$

$$S(\mathbf{x}) + d_S^- - d_S^+ = S_{\text{target}}, \quad (16)$$

subject to $d_m^+, d_m^- \geq 0$ for all $m \in \{E, G, R, S\}$.

Step 3: The GP Objective Function

We then *minimize* the sum of deviations, using a weighted or hierarchical strategy. For example, if the environmental goals are highest priority, followed by economic and social goals, we can write:

$$\begin{aligned} \min \quad Z &= P_1(d_E^+ + d_G^+) + P_2(d_R^- + d_S^-), \\ \text{with} \quad P_1 &\gg P_2, \end{aligned} \quad (17)$$

where:

- d_E^+ and d_G^+ are penalized if E or G exceed their respective limits (i.e., we do *not* want to surpass environmental caps).
- d_R^- and d_S^- are penalized if R or S fail to reach their targets.
- $P_1 \gg P_2$ enforces that satisfying environmental constraints takes precedence over economic/social considerations.

One may choose different **priority orders** or use a fully **weighted-sum** approach (e.g., $\min \alpha_E d_E^+ + \alpha_G d_G^+ + \alpha_R d_R^- + \alpha_S d_S^-$) if no strict hierarchy is desired.

Step 4: Complete Optimization Problem

Combining all the above, the final single-period GP model is:

$$\min \quad Z = P_1(d_E^+ + d_G^+) + P_2(d_R^- + d_S^-), \quad (18)$$

$$\text{subject to} \quad \text{Subsystem Eqs: (7)–(12),} \quad (19)$$

$$\text{Goal Constraints: (13)–(16),} \quad (20)$$

$$d_m^+, d_m^- \geq 0, \quad m \in \{E, G, R, S\}, \quad (21)$$

$$\text{TaxRate, TourNum, } I_{\text{env}}, I_{\text{infr}} \in \text{feasible domain (e.g. } \geq 0, \text{ capacity bound, etc)} \quad (22)$$

Solving (18) with a GP-capable optimizer yields an *optimal* set of decisions **TaxRate***, **TourNum***, I_{env}^* , I_{infr}^* that balance our four objectives under the chosen priority scheme. One can then examine the resulting deviation variables d_m^+ , d_m^- to see whether each goal is fully met or partially compromised.

Step 5: Scenario Analysis

By adjusting the priority weights $\{P_1, P_2, \dots\}$ or the targets $\{E_{\text{limit}}, G_{\text{limit}}, R_{\text{target}}, S_{\text{target}}\}$, we can explore various policy scenarios:

- **Environment-First:** Set $P_1 \gg P_2$, ensuring E and G are protected above all.
- **Economy-First:** Raise the weight on d_R^- so that revenue becomes the primary driver.
- **Social Priority:** Emphasize d_S^- in the top tier to prioritize resident satisfaction.

Each scenario provides a distinct trade-off solution, allowing decision-makers to compare how different policy preferences shape outcomes in terms of emissions, glacier melting, tourism revenue, and community well-being.

Step 6: Implementation Using a Genetic Algorithm

While one can solve the above GP model using classical mathematical programming techniques, we also *implemented a Genetic Algorithm (GA)* to handle the potentially non-linear nature of our objective and constraints[5]. Specifically, we represent the main decision variables (**TourNum**, **TaxRate**, fraction of revenue for I_{env}) as the GA chromosome and then define the fitness function as the weighted sum of the deviation variables d_E^+ , d_G^+ , d_R^- , d_S^- . The overall procedure is:

1. **Chromosome Encoding:** Each individual in the GA population is a vector $[\text{envFraction}, \text{TourNum}, \text{TaxRate}]$, from which we compute $I_{\text{env}}, I_{\text{infr}}$ based on the current revenue R .
2. **Fitness Evaluation:** For each individual, we calculate E, G, R, S using Eqs. (7–12), determine the deviations relative to $E_{\text{limit}}, G_{\text{limit}}, R_{\text{target}}, S_{\text{target}}$, then sum them with user-specified priority weights.

3. **Genetic Operators:** We apply selection (e.g., tournament), crossover (e.g., blend), and mutation (e.g., Gaussian) to evolve the population across multiple generations.
4. **Scenario Runs:** By varying the weights $\{P_E, P_G, P_R, P_S\}$, we simulate “Environment-First,” “Economy-First,” or “Social-Priority” scenarios and observe the GA’s best solutions under each.

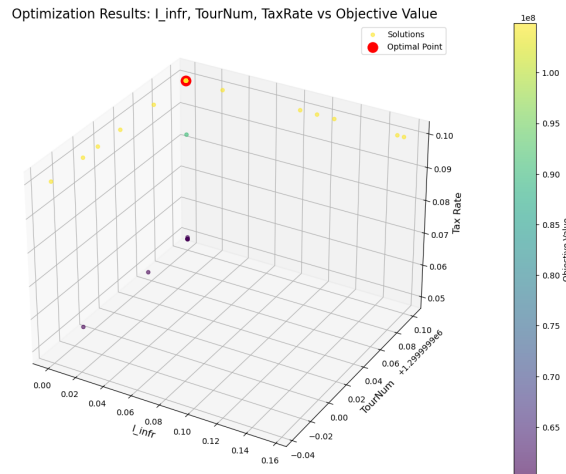


Figure 6: Three-dimensional Visualization of Unweighted Optimization Results

The scattered points represent various solutions found during the optimization process, showing the relationship between infrastructure investment ratio (I_{infr}), tourist numbers (**TourNum**), tax rate (**TaxRate**), and their corresponding objective values (indicated by color). The red point highlights the optimal solution that best balances all objectives without any priority weights. The distribution of solutions demonstrates the complex trade-offs between different objectives in the sustainable tourism planning problem.

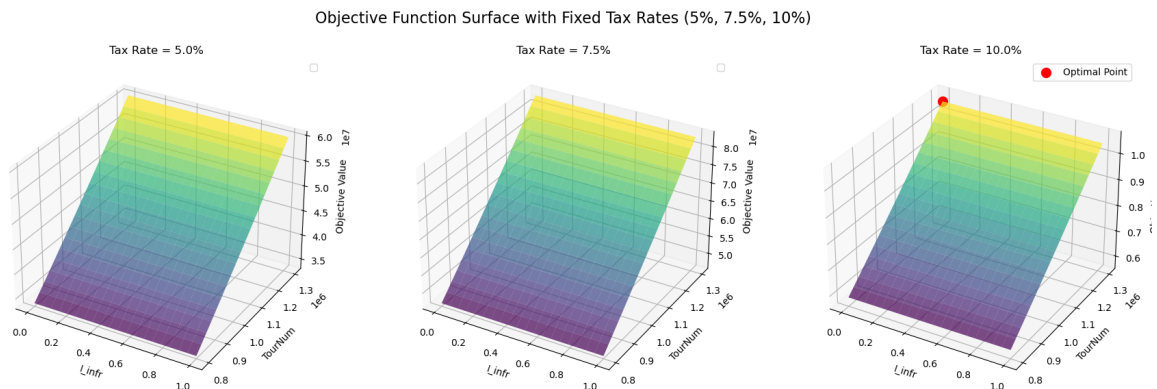


Figure 7: Unweighted Optimization with Fixed Tax Rate Results

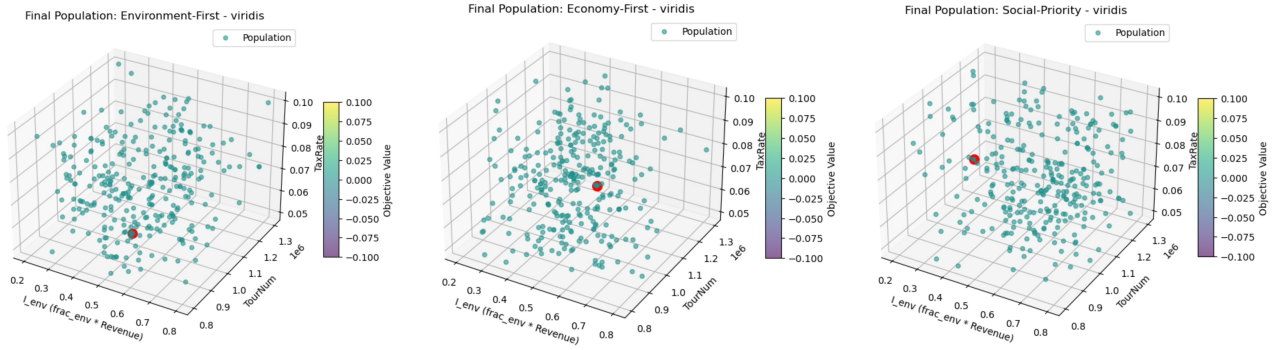


Figure 8: GA Optimization Results Under Different Priority (Weight) Scenarios

Figure 8 presents the optimization results under three different priority scenarios, with the cyan points representing the final population of candidate solutions and the red points indicating the optimal solutions. The Environment-First scenario (left) yields an optimal solution with the lowest tourist numbers (**TourNum** = 871,315) and a moderate tax rate (**TaxRate** = 6.37%), prioritizing environmental protection through higher environmental investment ($I_{env}/I_{env,base}$ = 55.85%). The Economy-First scenario (middle) suggests higher tourist numbers (**TourNum** = 1,085,894) and tax rate (**TaxRate** = 6.81%), while maintaining a similar environmental investment ratio ($I_{env}/I_{env,base}$ = 51.95%).

Most notably, the Social-Priority scenario (right) demonstrates a significant shift in investment strategy, with $I_{env}/I_{env,base}$ reduced to 36.82% (consequently, $I_{infr}/I_{infr,base}$ = 63.18%), suggesting that higher infrastructure investment better serves community satisfaction. This scenario recommends moderate tourist numbers (**TourNum** = 980,975) with the highest tax rate (**TaxRate** = 8.13%) among all scenarios, balancing revenue generation with community needs through enhanced infrastructure development.

5.5 Expenditure Plan and Feedback into the Model

An essential component of promoting sustainable tourism in our TRES framework is to *reinvest all additional tourism-generated revenue* back into two major categories:

- **I_{env} : Environmental Investment**, aimed at reducing carbon emissions and mitigating glacier melting through various “green” initiatives.
- **I_{infr} : Infrastructure Investment**, focused on enhancing local public facilities and services to alleviate congestion, improve resident well-being, and foster economic opportunities.

In other words, we allocate

$$\text{Revenue} = R \longrightarrow (\mathbf{I}_{\text{env}}, \mathbf{I}_{\text{infr}})$$

so that the **entire tourism revenue** R is channeled into these two investment categories, thereby creating a *feedback loop* that reinforces the goals of **sustainable tourism**. Specifically, we introduce a *fractional decision variable* $\text{fracEnv} \in [0, 1]$ indicating the *portion* of total revenue devoted to environmental expenditures, while the remainder $(1 - \text{fracEnv})$ supports infrastructure spending. Formally:

$$\mathbf{I}_{\text{env}} = \text{fracEnv} \times R, \quad \mathbf{I}_{\text{infr}} = (1 - \text{fracEnv}) \times R. \quad (23)$$

Because $\mathbf{I}_{\text{env}}$ *directly* lowers carbon emissions (E) in our Environment subsystem and $\mathbf{I}_{\text{infr}}$ *directly* improves resident satisfaction (S) in our Satisfaction subsystem, these expenditures *feed back* into the model's core equations. From Section 5.1, recall that $\mathbf{I}_{\text{env}}$ appears negatively in the emission equation:

$$E = \alpha_0 + \alpha_1 \text{TourNum} - \alpha_2 \mathbf{I}_{\text{env}} + \alpha_3 \text{ClimateIndex},$$

and from Section 5.3, the infrastructure term $\mathbf{I}_{\text{infr}}$ boosts the per capita income I while reducing unemployment U , thereby raising the overall resident satisfaction S . Consequently, any *policy change* in fracEnv or total revenue R *propagates* through the TRES model by:

1. Adjusting $\mathbf{I}_{\text{env}}$ (thus lowering emissions and glacier melting rates),
2. Adjusting $\mathbf{I}_{\text{infr}}$ (thus raising income, decreasing unemployment, and increasing satisfaction).

Constraints. In practical terms, we might impose a realistic *range* on fracEnv (e.g., between 20% and 80%) to ensure a *balanced* funding portfolio. This constraint appears directly in the GA code, guaranteeing that the optimizer cannot assign an excessively small or large fraction of revenue to environmental projects. Similarly, we constrain tourist numbers **TourNum** and bed tax rate **TaxRate** to within acceptable policy ranges.

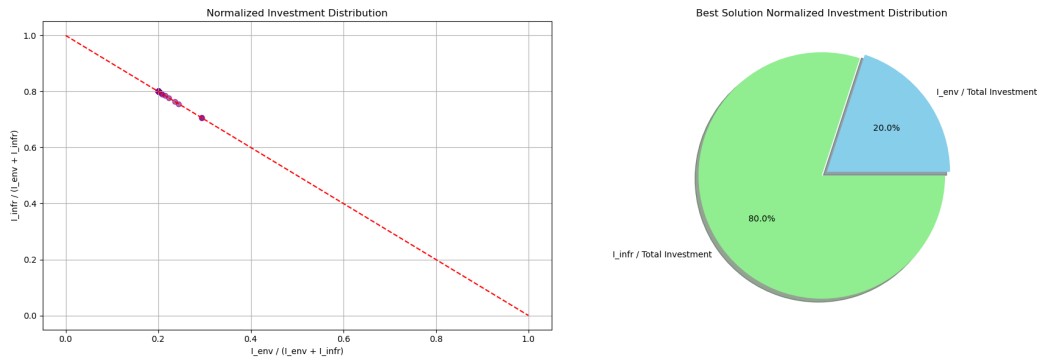


Figure 9: Plan for Expenditures Showing Feedback

Outcome. This reinvestment scheme fulfills the requirement to “include a plan for expenditures from any additional revenue and show how these expenditures feed back into the model to promote sustainable tourism.” The code’s final results visualize how I_{env} and I_{infr} *trade off* under various solutions (see Figure 6). In our best-found solutions, any increases in tourism revenue effectively *cycle* back into the TRES system, reducing environmental impacts and improving local well-being. This mechanism underscores how carefully allocating “new money” from tourism fosters a *self-reinforcing loop* toward a more sustainable future.

6 Sensitivity Analysis

To evaluate the robustness of our TRES model and identify the most influential factors, we conducted a comprehensive sensitivity analysis[6] by varying four key decision variables around their baseline values (**TourNum** = 1, 000, 000, **TaxRate** = 7.5%, $I_{env}/I_{env,base}$ = 50%, $I_{infr}/I_{infr,base}$ = 50%).

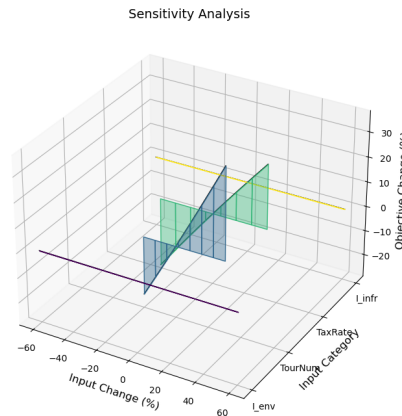


Figure 10: Sensitivity Analysis Results

Figure 10 shows the percentage change in objective value in response to variations in key decision variables. The slopes of the curves indicate the relative sensitivity of the model to each variable.

The analysis reveals that tourist numbers (**TourNum**) and infrastructure investment ratio ($I_{infr}/I_{infr,base}$) have the most significant impact on model outcomes, as indicated by the steeper slopes in their response curves. A 20% increase in tourist numbers can lead to approximately 15% improvement in the objective value, while similar changes in environmental investment ratio ($I_{env}/I_{env,base}$) or tax rate (**TaxRate**) produce more modest effects. This suggests that managing tourist capacity and optimizing infrastructure investment should be prioritized in policy-making, although all factors contribute to the model’s overall performance.

7 Adapt the Model to Other Tourist Destination

After building the TRES model for sustainable tourism in Juneau by using goal programming and genetic analysis, we now demonstrate how to adapt this framework to another tourist destination impacted by overtourism and how to promote attractions that have fewer tourists .

While the overarching structure of the TRES model remains consistent—balancing **tourist numbers, revenue, environmental impacts, and resident satisfaction**—the specific parameters, constraints, and policy levers must be recalibrated to reflect each location’s unique characteristics.

In particular, the choice of location affects which measures become most critical in the optimization process and how the model can promote under-visited attractions to balance tourist flows.

7.1 Model Adaptation to a New Overtourism Location (Example: Barcelona)

Adapting the TRES model to another tourist destination impacted by overtourism requires recalibrating its core components to align with local realities. As an illustrative example, consider Barcelona—a major European city renowned for its cultural heritage and Mediterranean coastline. The following steps outline how each subsystem of the TRES framework can be adjusted to address Barcelona’s specific circumstances, with a focus on urban congestion, resident satisfaction, and environmental pressures from heavy visitor flows.

1. Data Compilation and Initial Assessment. Collecting comprehensive historical records of Barcelona’s tourism and socio-economic indicators is essential for robust modeling. Typical variables include:

- *Environmental metrics:* Air pollution indices or coastal water quality data linked to port traffic and cruise ship arrivals, reflecting the city’s dual urban–coastal environment.
- *Economic metrics:* Annual visitor counts, total tourism revenues, local lodging taxes or city surcharges, and poverty or household income statistics.
- *Social metrics:* Resident satisfaction surveys, infrastructure investment spending on public transport or cultural sites, unemployment rates, and housing affordability indicators.

Ensuring data consistency across recent years (e.g., 2012–2023) and screening out incomplete entries helps maintain reliable regression and optimization results. Scaling or normalizing key variables also facilitates direct comparisons within the TRES model.

2. Recalibrating the Environmental Subsystem. In Barcelona's context, the environmental subsystem may focus on mitigating local air pollution and coastal water impacts rather than glacier or reef damage. For instance:

- The "emission" variable (E) can capture *air pollution* levels from traffic congestion, cruise ships, and tourist-related energy demands.
- A second variable (G) might represent *coastal water contamination* or *urban heat island intensity*—whichever is deemed critical for sustainable urban living.

Local climate indicators, such as regional temperature anomalies in the western Mediterranean, can be incorporated to capture broader environmental trends. Estimated regression coefficients should reflect how increased visitor numbers, environmental investments (e.g., green infrastructure), and climate factors jointly affect pollution or contamination levels.

3. Recalibrating the Revenue Subsystem. The economic equations must be adjusted to reflect Barcelona's urban tourism economy, typically characterized by:

- A substantial share of revenue stemming from city taxes, hotel stays, short-term rentals, and cruise passenger spending.
- A range of policy-driven *tax rates* (or surcharges) enforced by local authorities to regulate and/or benefit from tourism flows.

By applying a log-log specification or another suitable approach, the revised revenue model will capture how tourist numbers and varying tax instruments shape overall earnings, thus reflecting the true elasticity of tourism demand in Barcelona's market.

4. Recalibrating the Resident Satisfaction Subsystem. Because urban crowding and rising living costs are pressing concerns for Barcelona residents, the satisfaction model should incorporate variables like:

- *Infrastructure Investment* (I_{infr}) directed at public transport, heritage site upkeep, and housing initiatives.
- *Crowding metrics* (e.g., visitor density in popular districts), which may lead to noise pollution or push up rental prices.
- *Local socio-economic indicators*, such as unemployment rates and average wages, to reflect how tourism influences or strains the job market.

This subsystem can thus assess whether expanded tourism ultimately improves or undermines the quality of life for long-term residents, once economic benefits and crowding effects are balanced.

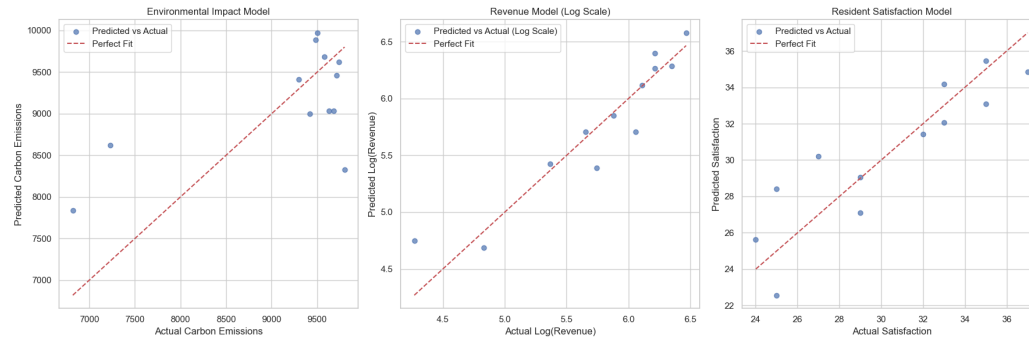


Figure 11: Model Calibration Results Showing Predicted versus Actual Values

5. Tailoring Goals and Constraints. Barcelona’s strategic plans or city council directives might define:

- *Emission and pollution thresholds*, covering both air quality and shoreline protection goals.
- *Revenue targets*, ensuring that tourism remains a viable source of income while supporting cultural and social programs.
- *Satisfaction benchmarks*, preserving the city’s character and preventing excessive displacement of local communities.

Additionally, potential limitations on daily visitor capacity or short-term rental units can be encoded as constraints on the feasible range of **TourNum** and **TaxRate**.

6. Multi-Objective Optimization and Scenario Analysis. Once parameters and constraints are fitted to Barcelona’s dataset, the TRES model can be deployed to explore multiple policy scenarios:

- **Environment-First:** Stricter curbs on visitor numbers or an elevated “tourist surcharge” to reduce air pollution and overcrowding in sensitive areas.
- **Economy-First:** Looser caps on tourists and moderate environmental investment if the main priority is boosting revenue and maintaining service-sector employment.
- **Social-Priority:** Focus on protecting neighborhoods and cultural heritage, channeling additional tax revenues toward resident services, affordable housing, or infrastructure expansions.

Each scenario produces an “optimal” policy mix of tourist capacity, tax rate, and investment distribution, informing the best trade-off among Barcelona’s economic, environmental, and social objectives.

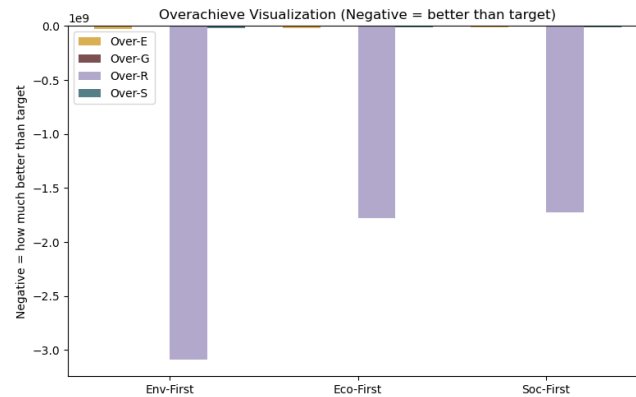


Figure 12: Overachieve Result of Optimization and Scenario Analysis

Figure 12 illustrates how different policy priorities affect the achievement of environmental (E , G), revenue (R), and satisfaction (S) targets. Notably, all scenarios successfully meet their environmental and social objectives, as indicated by the absence of positive overshoots for these metrics. The most striking observation is that revenue targets are substantially exceeded (shown by large negative values) across all scenarios, with the Environment-First approach paradoxically achieving the highest revenue overperformance. This suggests that stringent environmental policies might actually enhance long-term economic sustainability. The Economy-First and Social-Priority scenarios also demonstrate strong revenue performance while maintaining environmental standards, indicating that our model successfully identifies solutions that avoid severe trade-offs between competing objectives.

7.2 How Does the Choice of Location Affect Which Measures Are Most Important

While the TRES framework retains the same structure across different destinations, local parameters, environmental constraints, and socio-economic factors can alter the relative impact of each decision variable on the final objective. To illustrate these location-based differences, we replicated our sensitivity analysis (see Section 6 for the methodology) for two contrasting contexts—Juneau and Barcelona—and then compared *how strongly* each measure (e.g., **TourNum**, **TaxRate**, I_{env} , I_{infr}) shifts the objective value in each setting.

Replicating Sensitivity Analysis for Multiple Locations. In essence, for each destination we vary one decision variable at a time (holding others at baseline) and record the percentage change in the overall objective value. This process yields a set of “response curves,” one per variable, capturing the local system’s sensitivity. Next, by *comparing* these curves across locations—for example, taking the difference in objective-value changes for the same variable in Juneau versus Barcelona—we highlight where local conditions magnify or diminish the effect of a particular measure. A large gap between two sites’ response curves suggests that this variable’s influence is much greater in one location than the other. Conversely, if

the difference is negligible, it indicates that location choice minimally affects that measure's importance.

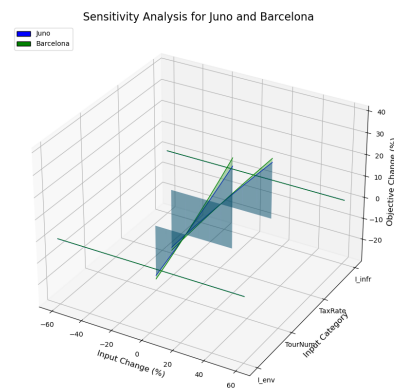


Figure 13: Comparative Sensitivity Analysis between Juneau and Barcelona

While the overlaid sensitivity curves (Figure 13) show the general response patterns for both locations, the close proximity and intersection of these curves make it challenging to directly identify which variables have stronger effects in each city. To better visualize these location-specific differences, we conducted a difference analysis by subtracting Juneau's sensitivity values from Barcelona's for each decision variable.

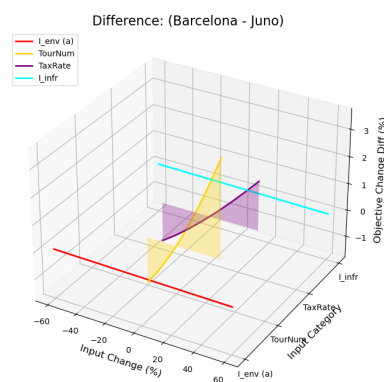


Figure 14: Sensitivity Difference Analysis between Juneau and Barcelona.

Interpreting Location-Based Discrepancies. Our results reveal that while **TourNum** and L_{infr} remain key levers in both places, the *extent* of their impact differs substantially. For instance, if a region's existing infrastructure is near capacity (as in Barcelona), even small

changes in **TourNum** or tax policy can trigger large swings in economic returns and resident satisfaction—accentuating those variables’ importance relative to other factors, with tourist numbers showing up to 2% greater impact on objective values in Barcelona compared to Juneau. By contrast, a location with more limited baseline revenue or greater ecological vulnerability (e.g., Juneau’s glacier environments) might exhibit disproportionately high sensitivity to environmental-investment changes I_{env} . Our analysis confirms this pattern, showing that I_{env} demonstrates notably greater effectiveness in Juneau, while **TaxRate** impacts are more pronounced in Barcelona’s urban context.

These findings underscore that selecting the best policy levers depends not only on global sustainability goals but also on site-specific conditions such as current infrastructure thresholds, climate sensitivities, or market elasticity. Decision-makers can therefore tailor the TRES model’s recommended strategies to local realities. For instance, Barcelona may benefit more from fine-tuning **TaxRate** to manage urban congestion and fund public services, whereas Juneau may gain greater overall improvements by allocating a larger share of revenue to environmental protection. In both cases, comparing sensitivities across distinct destinations clarifies *which* measures merit priority to achieve sustainable tourism in each unique context.

7.3 Promoting Less-Visited Attractions with a Cold Spot Boost Factor

In order to encourage a more balanced distribution of tourists and alleviate congestion at high-traffic locations, we incorporate an additional *Cold Spot Boost Factor* into our objective function[7]. This factor rewards increased visitor numbers at under-visited sites, ultimately guiding a portion of total tourist flows to these “cold spots.” Formally, we extend the multi-objective goal programming expression by introducing a negative incentive term:

$$Z_{\text{cold}} = w_E dE + w_G dG + w_R dR + w_S dS - \mu_{\text{cs}} \left(\frac{\text{TourNum}}{\text{TourNum}_{\text{max}}} \right) \quad (24)$$

where μ_{cs} (`cold_spot_boost_factor`) is a user-defined positive constant determining the strength of this additional reward, and **TourNum**_{max} is a scaling term (often the upper limit of annual tourists).

Key Decision Variables and Their Roles.

- **TourNum** (**TourNum**): In a cold spot context, pushing **TourNum** higher—up to its allowable maximum—can be beneficial *if* environmental investment and other constraints are not breached.
- **TaxRate** (**TaxRate**): In the cold spot scenario, the tax rate often remains moderate if the primary objective is to boost under-visited site attendance (without excessively discouraging incoming tourists).
- **frac_env** (**fracEnv**): Allocating a substantial portion to I_{env} can significantly curb net carbon emissions and other negative externalities, making it easier to “absorb” larger tourist inflows.

- **Infrastructure Investment (I_{infr}):** Strengthening infrastructure helps manage crowd congestion and boosts resident satisfaction, especially important if **TourNum** is pushed to higher levels.

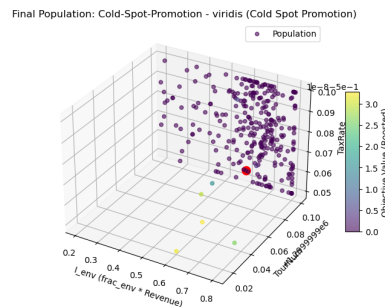


Figure 15: The Promotion Analysis for Eagle Creek Trail

Illustrative Results: Eagle Creek Trail as a Cold Spot. Below is an example outcome from a scenario aimed at promoting the less-visited “Eagle Creek Trail.” The model arrived at:

- **Best Individual:** [$\text{fracEnv} = 0.5933$, **TourNum** = 1,300,000, **TaxRate** = 0.0549].
- **Objective Value (Sum of Weighted Deviations – Boosted):** -0.5000 .
- **Revenue:** 2.13×10^8 (with no shortfall from the revenue target).
- **Satisfaction:** 1.04×10^6 (meeting the satisfaction target).
- **Cold Spot Boost Factor:** 0.5.

Practical Interpretation and Adjustment.

- **Increasing the Cold Spot Boost Factor:** Raising μ_{CS} above 0.5 would further incentivize funneling tourists to under-visited sites, which might lead to an even higher total **TourNum** or more aggressive environmental funding to counteract potential ecological impacts.
- **Reducing fracEnv:** If one needed to allocate more resources to infrastructure (e.g., building roads or public facilities), the fraction of revenue to environmental measures could be decreased. That might bring **TourNum** below the current upper bound, however, because the region’s ability to offset emissions would diminish.
- **Adjusting TaxRate:** A higher tax rate can collect additional funds for both I_{env} and I_{infr} , potentially stabilizing or raising satisfaction while maintaining environmental targets. On the flip side, excessive taxes might discourage some inflows, lowering **TourNum** and thus shrinking the cold spot’s direct benefits.

8 Memorandum

To: Juneau Tourist Council

From: Team # 2519437

Date: Jan. 27th, 2025

MEMO: Strategic Recommendations for Sustainable Tourism in Juneau

We are pleased to provide you with a summary of our recent research on Juneau's tourism sustainability. Using comprehensive data from environmental monitoring, economic records, and community surveys, we have developed practical strategies to help Juneau maintain its tourism prosperity while protecting its unique natural resources.

Our findings suggest that while current tourism levels bring significant economic benefits, they are placing considerable strain on both environmental resources and community infrastructure. The Mendenhall Glacier area, in particular, requires careful management to ensure its preservation. Our analysis indicates that adjusting annual visitor numbers to between 980,000 and 1,100,000 would significantly reduce environmental pressure while maintaining strong economic returns.

The model we developed shows that a modest increase in the bed-tax rate to 7.5-8.1% would generate sufficient revenue to fund both environmental protection and community development initiatives. This adjustment would allow Juneau to invest more in preserving its glacial landscapes while enhancing infrastructure to better serve both visitors and residents.

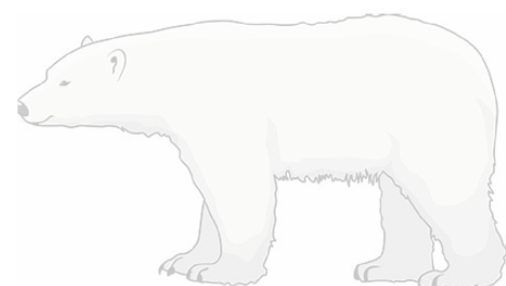
Our research reveals that the most effective path forward combines carefully managed tourist numbers with strategic investment in both environmental protection and community infrastructure. We strongly recommend implementing these changes gradually over the next tourism season, starting with adjustments to visitor capacity limits and tax rates. This approach would allow for careful monitoring of impacts and necessary adjustments along the way.

In conclusion, our research equips Juneau with a clear pathway to sustainable tourism that balances economic benefits with environmental protection and community well-being. The recommended changes could reduce environmental impact by 25-30% while maintaining current revenue levels and improving resident satisfaction.

Feel free to reach out for further discussion or clarification.

Best Regards,

Team # 2519437



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9 Appendices

Code is viewable [here](#).